Writing a Research Proposal

EEX6181-EEX6182

Types of Research Proposals

- **Undergraduate Research proposals**
- Postgraduate Research Proposals
- Research proposals for research grants

Content in a Research proposal

- Table of content, List of Figures and Tables
- Abstract
- Introduction
- Literature Review
- Research Methodology
- Time plan
- List of References

Introduction

Introduces your topic
Write your problem statement and the questions your research aims to answer
Provides context for your research
Can be a few paragraphs long
Must be simple and concise

Research Problem

- Problem identification
  - What are the gaps in current research ?
  - Are there demanding situations that demand immediate research?
  - Are you planning to introduce a new product, a mathematical model or a process?
  - Did you read adequate journal papers and other resources available?

Literature review

- Set the background for what has been researched on the topic so far
- Identify major themes and concepts
- Illustrate relationships between different ideas
- Show the gap in research and why particular topic is important
- introduce all the sources you plan to use in your research
  - This includes landmark studies and their data, books, and scholarly articles.

Literature Review contd.

- a literature review explore the collection of articles, conference papers, book chapters etc that you have chosen and,
  - explain how you're using them in **your** research

- A literature review is not a list of references

Research Design

- What is the type of research you will do ?
  - Are you collecting original data or working with data collected by other researchers?
- How will you collect data ?
- Whether you're doing experimental, correlational, or descriptive research ?
- What are the tools you will use to collect data ?
  - Will you be running experiments?
  - Conducting surveys?
  - Observing phenomena?
  - etc
- Use the most *effective methods for your specific research.*

Timeline for the Research Project

- Draw a chart to show how you will spend the year ahead with this research project
- From the month you started until now – you can write as conducting the Literature survey and writing the research proposal

Abstract

- Last section to be written but must be placed before the introduction
- One or two paragraphs, coherent and should be able to stand alone as one unit of information
- Covers full study – background, purpose, methods used, conclusion
- Write in simple English to be understood by a wider audience as well as domain experts
- Use passive voice, focus on issues rather than people

Thank you